

• • LIVE WEEKEND COHORT

Classical Mechanics Weekend

Newton to energy conservation — one focused weekend. Physical intuition first, equations second. By Day 2 you'll be solving problems you never thought you could.

PRICE	DURATION	LEVEL	FORMAT	SCHEDULE
₹1,999	2 Days · 12 hrs	Beginner	Live	Bi-weekly weekends



Prasant Mishra

Founder & Instructor, Qurious Academy

What is the Classical Mechanics Weekend?

Classical mechanics is the foundation of all physics. Before quantum mechanics, before relativity, before any of the exotic frontiers of modern physics — there are Newton's laws, kinematics, energy, and momentum. Mastering these isn't just useful for exams; it builds the physical intuition and problem-solving discipline that every serious physics student needs.

This weekend cohort covers the complete core of classical mechanics in two rigorous, accessible days. Day 1 is about motion and forces: kinematics (how things move), Newton's three laws (why they move), and the systematic problem-solving approach using free body diagrams. Day 2 adds energy and momentum: conservation laws that often give you elegant solutions to problems that would take pages with Newton's laws alone.

The teaching style is physical intuition first — we ask "what do you expect?" before we write an equation. This is the habit that separates students who truly understand mechanics from those who are just pattern-matching to formulae. Problem sets with full solutions are included for continued practice after the weekend.

IS THIS RIGHT FOR YOU?

Who Should Attend

✓ THIS IS FOR YOU IF...

- ✓ You're preparing for JEE, NEET, or university-level physics and want genuine conceptual clarity in mechanics — not just formula memorisation
- ✓ You've found mechanics problems confusing because you're not sure which equation to apply when
- ✓ You're a 11th or 12th grade student who wants to get ahead or catch up on classical mechanics before exams
- ✓ You studied mechanics previously and want a rigorous, systematic refresher that fills conceptual gaps

✗ THIS IS NOT FOR YOU IF...

- ✗ You have no algebra background at all — you need to be comfortable with equations, substitution, and basic trigonometry
- ✗ You want advanced topics like rigid body dynamics in depth, Lagrangian mechanics, or fluid mechanics — this weekend covers the core classical mechanics syllabus
- ✗ You're looking for university-level analytical mechanics with calculus throughout — this session uses calculus only where necessary and focuses on algebraic problem-solving

Prerequisites: Basic algebra and trigonometry (sin, cos, tan in right triangles). 11th-grade physics familiarity is helpful but not required — we start from kinematics definitions.

Two Days, Two Themes

DAY 1**Motion & Forces**

- ▶ **Kinematics — displacement, velocity, acceleration** — the language of motion; scalar vs. vector quantities; the four kinematic equations derived from first principles and their conditions of validity
- ▶ **Newton's three laws in depth** — not just statements but deep physical understanding: what inertia means, what force really is, why the third law pairs are never on the same object, and the common misconceptions that kill problem-solving
- ▶ **Friction, normal force & tension** — the contact forces that appear in almost every mechanics problem; static vs. kinetic friction, why the normal force is always perpendicular, tension in strings over pulleys
- ▶ **Problem-solving method: FBDs & systematic approach** — Free Body Diagrams as the universal problem-solving tool; a rigorous, repeatable framework for attacking any mechanics problem from first principles

DAY 2**Energy & Momentum**

- ▶ **Work, power & kinetic energy** — the work-energy theorem as an alternative to Newton's second law; when the energy approach gives you a simpler solution; power as rate of energy transfer
- ▶ **Conservation of mechanical energy** — potential energy (gravitational and spring), total mechanical energy, and when it's conserved; solving problems by tracking energy rather than forces
- ▶ **Momentum & impulse** — linear momentum as a conserved quantity; impulse-momentum theorem; the connection between force, time, and change in motion
- ▶ **Collisions — elastic & inelastic** — applying conservation of momentum and energy to collisions; perfectly elastic, perfectly inelastic, and the spectrum between; real-world collision problems
- ▶ **Rotational motion basics** — angular displacement, velocity, and acceleration; torque as the rotational analogue of force; moment of inertia and rotational kinetic energy

LEARNING OUTCOMES

What You'll Walk Away With

01 Solve mechanics problems from first principles

— not by recognising which formula to plug numbers into, but by building the solution from Newton's laws or conservation principles

02 Build genuine physical intuition

— the ability to predict qualitatively what will happen in a physical situation before solving it quantitatively; the hallmark of a strong physics student

03 Prepared for university physics or JEE/NEET mechanics

— the core classical mechanics content covered at a depth appropriate for competitive exam preparation and first-year university courses

WHAT'S INCLUDED

Everything You Get

✓ **Both day recordings** — full HD recordings of both sessions, shared within 24 hours

✓ **Problem sets with full solutions** — a comprehensive set of mechanics problems spanning both days, with complete worked solutions showing the full reasoning at each step

✓ **Learner's Certificate** — digital certificate of completion from Qurious Academy

LOGISTICS

Schedule & Pricing

SCHEDULE

Every other weekend (bi-weekly)

TIME

9:00 AM – 3:00 PM IST

DURATION

2 days · ~12 hours total

PRICE

❖ **Is this appropriate for JEE Main and JEE Advanced preparation?**

Yes — the core classical mechanics content is covered at a depth appropriate for JEE Main. For JEE Advanced, this weekend builds the foundation; you'll need additional problem-solving practice at the Advanced level.

❖ **What if I'm already comfortable with some topics?**

The first few topics in each day move quickly for students with prior exposure. The problem-solving method and energy/momentum sections tend to give even well-prepared students new perspectives.

❖ **Will I need a calculator?**

Yes — a scientific calculator is needed for numerical problems. A graphing calculator is helpful but not required.

About Prasant Mishra



Prasant Mishra

Founder & Instructor — Qurious Academy

GenAI Solutions Architect

15+ Yrs Teaching

2500+ Students

60+ Projects Delivered

Prasant has taught physics to students preparing for JEE, NEET, and university-level courses over his 15+ years of teaching. His approach to mechanics is rooted in the conviction that physical intuition is not a gift — it's a skill that's built through the habit of predicting before calculating, and checking whether your answer "makes sense" before moving on.

He's particularly effective at identifying and dissolving the conceptual misconceptions that cause students to fail at mechanics — the kinds of errors that can't be fixed by doing more problems without first changing the underlying mental model.

QuriousAcademy

Focused, live learning for curious minds. No fluff — just the concepts that matter.

Classical Mechanics Weekend

Live Cohort · Bi-weekly weekends · 9 AM–3 PM IST

₹1,999

Enroll at quriousacademy.com

or visit quriousacademy.com/courses

Questions? Reach us at hello@quriousacademy.com

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